

A-level

# ENVIRONMENTAL SCIENCE

## Questions by topic: The circular economy, biocapacity & ecological footprints - Set 1

### Mark scheme

AQA spec 3.6.4; Sustainability; textbook pp. 374-389

### Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

**Textbook references:** every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

| Question | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Marks |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 01.1     | <ul style="list-style-type: none"> <li>• Cycling of materials (like biogeochemical cycles) - recycling, re-use</li> <li>• Energy from renewable sources</li> <li>• Human activities should support ecosystems / conserve biodiversity</li> <li>• Separation of technical and biological materials; connected systems where the waste of one process feeds another (industrial symbiosis)</li> <li>• Diverse systems are more resistant to change</li> <li>• Design of products for extended use and end-of-life re-use; optimum rather than maximum production; technologies that improve efficiency and enable dismantling</li> </ul> | 6     |
| 01.2     | <ul style="list-style-type: none"> <li>• Natural systems over-produce so the surplus supports the processes the system relies on: plant products feed pollinators, seed dispersers, decomposers, mycorrhizae</li> <li>• Agroecosystems aim for maximum harvested yield, removing the surplus that would maintain soil biota, pollinators and pest predators, so they need artificial inputs</li> </ul>                                                                                                                                                                                                                                 | 2     |
| 01.3     | <ul style="list-style-type: none"> <li>• Costs of environmental measures could put companies in one country at a disadvantage against competitors in countries without them</li> <li>• A common approach gives a level playing field, coordinated monitoring and support for less successful countries</li> </ul>                                                                                                                                                                                                                                                                                                                      | 2     |
|          | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 374-375; AQA spec 3.6.4.1.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |       |

| Question | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Marks |
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| 02.1     | <p><b>Levels of response</b></p> <p><b>Level 3 (7-9 marks):</b> Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached.</p> <p><b>Level 2 (4-6 marks):</b> Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported.</p> <p><b>Level 1 (1-3 marks):</b> Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation.</p> <p><b>0 marks:</b> No relevant content.</p> <p><b>Indicative content</b> (credit other relevant, accurate points)</p> <ul style="list-style-type: none"> <li>• Water: catchment management, conservation, metering, greywater, re-use of treated effluent, aquifer recharge, sustainable urban drainage - largely cyclical already; limits in arid regions</li> <li>• Minerals: exploiting low-grade ores with low-temperature processes (bioleaching), extending product lifespans, repair / re-use / repurpose / recycle, design for disassembly, reduced packaging; limits: dispersal, alloys, energy of collection</li> <li>• Waste management: repair, re-use, refurbish, recycle, composting and anaerobic digestion returning nutrients, minimal landfill, energy from residual waste; EU targets on packaging, batteries, WEEE</li> <li>• Pollution control: shift from post-production treatment to non-release - fuel desulfurisation not FGD, fuel cells / electric vehicles not combustion, industrial symbiosis (Kalundborg: waste heat, gypsum, fly ash, yeast used by neighbours); limits for legacy pollution</li> <li>• Energy: renewables, storage, energy density technologies, low-temperature manufacturing, CHP; limits of intermittency, transport fuels</li> <li>• Food: natural processes for nutrients (legumes, organic matter), pest control (IPM), soil biota, agri-environment schemes, IMTA aquaculture; tension with maximum yield for a growing population</li> <li>• Land uses supporting ecosystems: urban green infrastructure, gardens, permeable surfaces, habitat in redeveloped docks / lagoons</li> <li>• Judgement: high applicability in some sectors (water, waste), partial in others (minerals, energy for transport); needs policy support</li> </ul> | 9     |
| 02.2     | <ul style="list-style-type: none"> <li>• Locating industries so that the wastes of one (heat, gases, by-products) become the raw materials of another, mimicking natural cycles (eg Kalundborg eco-industrial park)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 1     |
|          | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 375-379; AQA spec 3.6.4.1.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |       |
| 03.1     | <ul style="list-style-type: none"> <li>• Ecological footprint: the area of the Earth's surface (global hectares) needed to provide the resources used by a person / population / activity and to absorb the wastes produced</li> <li>• Biocapacity: the biological productivity of an area (forests, cropland, grazing, fisheries, built land) - its ability to supply resources and absorb wastes, expressed in global hectares</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2     |

| Question | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Marks |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 03.2     | <ul style="list-style-type: none"> <li>• A hectare of land / sea with the world-average biological productivity</li> <li>• Biocapacities of real areas vary widely so a standard unit allows footprints and biocapacities of different places to be compared</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 2     |
| 03.3     | <ul style="list-style-type: none"> <li>• <math>8.1 / 1.7 = 4.8</math> (accept 4.7-4.8)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 1     |
| 03.4     | <ul style="list-style-type: none"> <li>• Higher HDI is associated with a larger footprint (USA 8.1 vs Bangladesh 0.8), though the UK achieves the same HDI as the USA with about half the footprint</li> <li>• Global footprint (2.8) exceeds global biocapacity (1.7) so humanity uses resources faster than they regenerate and CO<sub>2</sub> accumulates (overshoot)</li> <li>• Earth Overshoot Day is the date each year when a year's worth of resources / waste absorption has been used up; it falls earlier each year (8 August in 2016)</li> </ul>                                                                                                                                                                                         | 3     |
| 03.5     | <p>Any 2 from:</p> <ul style="list-style-type: none"> <li>• Renewable energy and energy conservation (carbon footprint is the largest component)</li> <li>• Reduced meat consumption / food waste; local seasonal food</li> <li>• Circular economy: repair, re-use, recycling; longer product lives</li> <li>• Public transport, efficient buildings</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                      | 2     |
|          | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 379-380; AQA spec 3.6.4.2.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |       |
| 04.1     | <ul style="list-style-type: none"> <li>• All greenhouse gases expressed as CO<sub>2</sub> equivalent (allowing for different warming strengths)</li> <li>• Direct emissions from fuel combustion plus indirect: manufacture of components, maintenance items, final disposal (life-cycle analysis)</li> <li>• A fair share of the infrastructure: roads, car parks, signals, street lighting</li> </ul>                                                                                                                                                                                                                                                                                                                                              | 3     |
| 04.2     | <ul style="list-style-type: none"> <li>• Compares the populations of over 18 000 populations of 3700 vertebrate species (terrestrial, freshwater, marine) with 1970 levels</li> <li>• Published in the Living Planet Report as a measure of the state of the planet's ecosystems / biodiversity loss over time</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                            | 2     |
| 04.3     | <ul style="list-style-type: none"> <li>• Hardy, disease-resistant Boran cattle; regular tick dipping reduces ticks for wild herbivores too</li> <li>• Mobile bomas (night enclosures) protect cattle from lions and leave dung that enriches soil and plant growth</li> <li>• Electric fences keep game in and poachers out; biological corridors let animals (except rhinos) move to other reserves; problem elephants relocated</li> <li>• Monitoring: radio-collared lions, drones, camera traps, track surveys; endangered species (Grevy's zebra, northern white rhino) protected</li> <li>• Community links: ecotourism income funds health, education and water projects; local employment; cattle breeding help for local farmers</li> </ul> | 5     |
|          | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 380-383; AQA spec 3.6.4.2.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |       |
| 05.1     | <ul style="list-style-type: none"> <li>• Diversity: HEP (wet winters), solar (summer) and wind peak at different times so supply is more reliable - diverse systems resist change</li> <li>• No single source could meet demand all year; battery storage on the island grid smooths fluctuations</li> <li>• Calm, dry, dull periods still leave gaps so diesel generators provide security (fossil fuel use minimised, not yet eliminated)</li> </ul>                                                                                                                                                                                                                                                                                               | 3     |

| Question    | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Marks |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <b>05.2</b> | <p><i>Any 3 from:</i></p> <ul style="list-style-type: none"> <li>• Community ownership (Isle of Eigg Trust with Scottish Wildlife Trust) managing the island for people and wildlife</li> <li>• Wood biomass heating; insulation such as straw-bale housing; some electric vehicles</li> <li>• Habitat protection and creation: fencing rare plants from grazing, planting indigenous woodland (rowan, birch, willow); eagles and hen harriers nesting</li> <li>• Recyclables shipped off; garden / food waste composted; low-impact food production; beach litter collection</li> </ul> | 3     |
| <b>05.3</b> | <ul style="list-style-type: none"> <li>• Small groups can organise, agree and act quickly; decisions benefit people who see the results directly</li> <li>• Resources and wastes are local and visible so feedback is immediate, unlike cities supplied from far away</li> </ul>                                                                                                                                                                                                                                                                                                         | 2     |
| <b>05.4</b> | <ul style="list-style-type: none"> <li>• Renewable energy is abundant but geographically and temporally variable; a continental grid moves electricity from areas of surplus (desert solar, offshore wind) to areas of demand</li> <li>• DC grids do not need every generator synchronised in phase as AC grids do, so coordination across many countries is easier</li> </ul>                                                                                                                                                                                                           | 2     |
|             | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 384-389; AQA spec 3.6.4.                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |       |